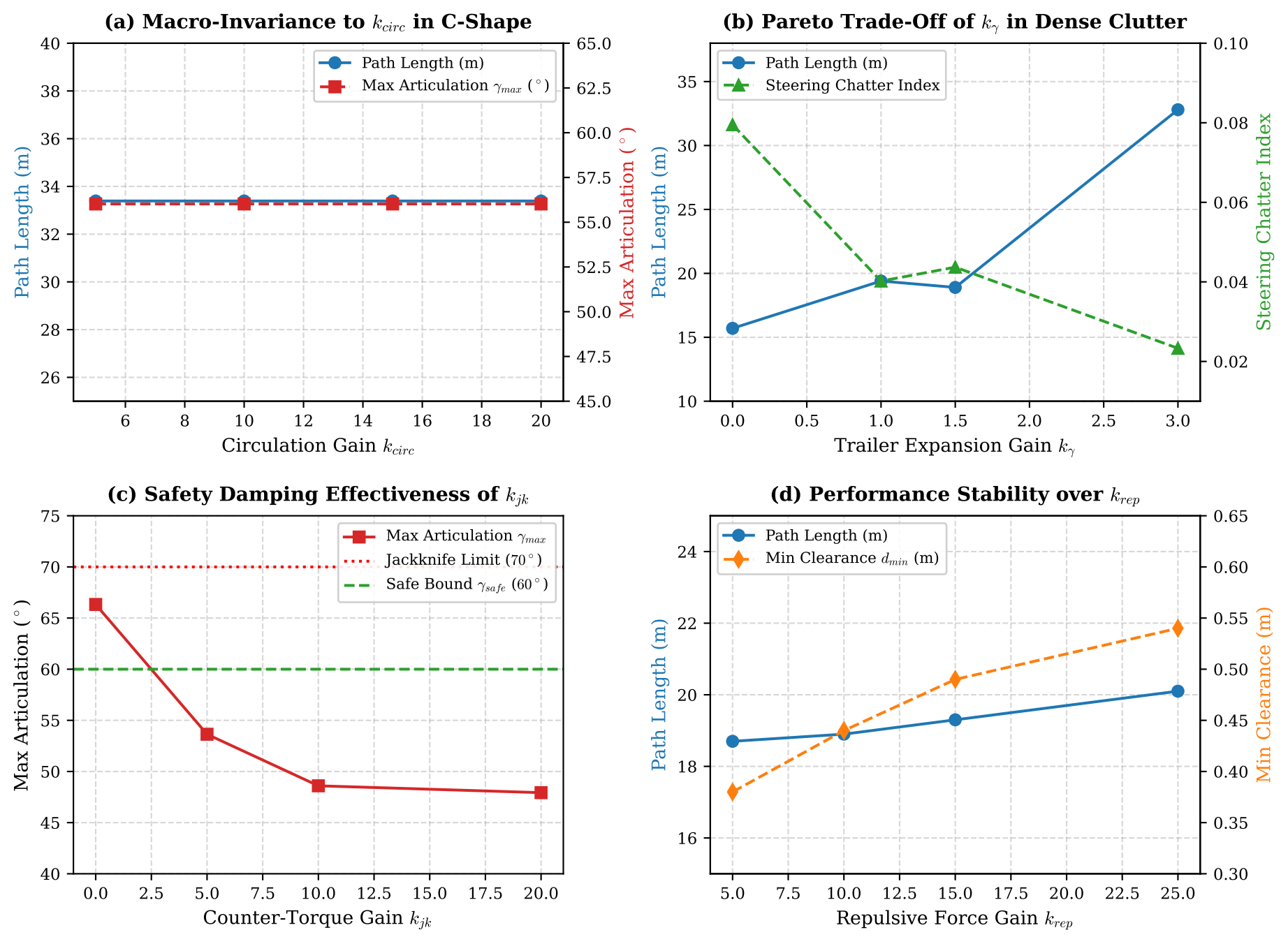
(a) Macro-Invariance to $k_{circ}$ in C-Shape
Path Length (m)
Max Articulation $\gamma_{max}$ (°)
Path Length (m)
Max Articulation (°)
Circulation Gain $k_{circ}$
(b) Pareto Trade-Off of $k_\gamma$ in Dense Clutter
Path Length (m)
Steering Chatter Index
Path Length (m)
Steering Chatter Index
Trailer Expansion Gain $k_\gamma$
(c) Safety Damping Effectiveness of $k_{jk}$
Max Articulation $\gamma_{max}$
Jackknife Limit (70°)
Safe Bound $\gamma_{safe}$ (60°)
Max Articulation (°)
Counter-Torque Gain $k_{jk}$
(d) Performance Stability over $k_{rep}$
Path Length (m)
Min Clearance $d_{min}$ (m)
Path Length (m)
Min Clearance (m)
Repulsive Force Gain $k_{rep}$